Week 5: The Avellaneda-Stoikov Engine

Implementation, calibration, and backtest – Milestone 1

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Learning Objectives

- Implement the quotes as a clean, tested PyTorch engine
- Calibrate the order-flow intensities by maximum likelihood
- Build an event-driven backtest and benchmark on Coincall data

Lecture 1: From Closed Form to Code

- Engine architecture; state management from the L2 book
- Mapping continuous-time quantities to discrete events
- Calibrating lambda(d) = A exp(-kappa d) from fill data
- Pitfalls: units, cash-process sign, inventory indexing

Lecture 2: Calibration & Validation

- Maximum-likelihood fit of A and kappa with goodness-of-fit
- The backtest harness: replay, fill simulator, P&L attribution
- Validate the P&L distribution against theory
- Reproducibility under a fixed seed

Milestone 1 Deliverable

- Python/PyTorch Avellaneda-Stoikov package on one linear asset
- Calibration + simulation + one-month BTC-PERP backtest
- Acceptance: quotes match closed form; P&L within +/-50% of reference
- pytest coverage; report <= 10 pages